Indian Institute of Management Calcutta

Programme: MBA

Name of the Course: **Economics of Climate Change: Implications for Financial Sector (* this would be cross listed with Finance and Economics)**

Core / Elective ……Elective…….. Cap (for Elective) ……none….

Credit: …1.5….

Academic Year & Term: …2026-27, VI

Course Coordinator: Avijit Bansal…….

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Instructor: Dhruba Purkayastha (Sessions 3-5, 7-8)

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Introduction to Course /Course Description

This course attempts to explain implications of climate change for the financial sector in India and globally.

It starts with the economic underpinnings of climate change and its implications on banks, financial institutions and capital markets and then discusses how climate risks affect financing and investment

decisions; Also discusses issues in financing of green investments (Green finance) and possible role of financial sector regulation for greening finance and managing climate change.

Course Objective and Key-takeaways from the course

The main objective of the course is to provide students with an understanding of the current theoretical and conceptual framework on implications of climate change (climate risks) for the broader economy, and then specifically the financial sector. To start with the students will apply microeconomic concepts of externalities and public goods to comprehend the failure of the financial sector to provide adequate financial flows for climate change related investments. Students will then be exposed to credit markets and the traditional financial risk management framework to understand climate change as a financial risk. Students will then be exposed to what climate finance entails and concepts of public finance, blended finance and institutional mechanisms for financing climate investments.

Learning Goals

1. Possess state-of-the-art knowledge of theory and practice in all functional fields of management and the ability to think critically, and apply them to diagnose and find solutions to organizational problems, even in unfamiliar or uncertain situations

3. Develop a global perspective that is built upon knowledge of economic, technological, regulatory and political forces that are shaping a new international order in the 21st century

International Components in your course (if any)

The course is about global financial markets and how they can address climate change. Starting from the political economy of climate change to global standards and regulations for financial markets, the entire course transcends local and national boundaries

Pre-requisites

Microeconomics, Risk Management, Corporate Finance

Required Textbook(s)

A Handout with relevant readings and papers will be distributed

Recommended Textbook(s) Handbook of Sustainable Finance – Theirry Roncalli (S & P)

Course Pack Distribution to students

Course material will be issued from the Programme office

Technology enabled learning component for your course

Online Course Management (moodle)/course web

Session Plan

S No.	Topic	Intended learning outcome	References	Instructor
1	Intro to Climate Change	The Science behind the Phenomenon, Why scientists believe the Earth's weather system is changing, how these changes affect natural systems, human society and the economy, Why, as a result of changes in the natural environment, in regulation, and in customer behaviour, this is relevant for business	South and Alpay (2026)The Belém Reckoning: Can the UNFCCC 'COPe' with Climate Change Effectively?, Climate and Energy An Introduction to Climate Change	RS
2	Environmental Economics	Review of positive and negative externalities, interventions required, market solutions, regulation and standards, public policy issues, economic implications of climate change for firms, review of IS-LM models and impact of monetary markets	Baomin Yang (2025), Current Development and Practice: The Economics of Climate Change, chapter in The Carbon Emission Liability Mechanism: A New Solution for Global Carbon Neutrality , Springer	RS
3	Climate Policy - Global and India	Global compacts, UNFCCC, From Kyoto to Paris and subsequent CoPs and NDCs (add NAPs), BTRs - Article 6 , national commitments towards mitigation and adaptation, progress and failures in the last 10 years, Trade and Climate and International Climate Finance, North-South Divide, India's Climate Policy, NDC, BRSR, NZT, Green Bonds. Solar Programs EV Programs , Indian Carbon Markets	Carbon pricing leadership coalition – What is carbon pricing, https://www.carbonpricingleadership.org/what **Sharma (2023), Political Economy of Climate Change, Economic and Political Weekly, Vol. 58, Issue No. 25-26, 24 Jun, 2023**	DP
4	Implications of Climate Change on Banking and Finance - Banks/NBFCs, Asset Managers, Insurance	Climate/Green/Sustainable Finance: Pricing credit, capital provisioning, climate-related financial risk assessments, and instruments to incentivize carbon-neutral or low-carbon investments . How climate change affects Finance?	Green Finance–Leading Initiatives and Research from international financial institutions, India. Net Zero Transition Finance Report from Niti Aayog	DP
5	Green and Climate Finance	Issues of financing green and climate investments - clean energy, energy efficiency, clean transport, industrial decarbonization, adaptation and resilience measures, carbon removal etc.	IDFC Green Finance Report. 2025 NGFS (Network for Greening Financial System) –Progress Reports Other contemporary readings	DP
6	Implication for Capital Markets	Role of Impact investing in capital allocation	Case: The Rise Fund: TPG Bets Big on Impact (9-318-041)	AB
7	Public Finance - Green Budgeting	Use of public finance (development finance, philanthropy, climate	UNEP Finance Initiative – Changing	DP

	Blended finance	capital) in mitigating and adapting to climate change	Course, May 2019 Other contemporary readings	
8	Policy and Institutional Support for Financing Climate Investments	Instruments available to the government to support investments (subsidies, public procurement, regulations), risk mitigation, role of the financial sector regulators, Implications for Monetary Policy and Central Banking role	Delivering a Sustainable Financial System in India- FICCI. 2018. Framework for Climate Finance Taxonomy in India, Ministry of Finance GoI Other contemporary readings	DP
9	Climate Change as a Financial Risk	Review of understanding on credit markets and risks, the implications of climate change on physical and transition risks, How climate change impacts financial services, including a break down by sector, i.e. banking, insurance, and asset management	Book Chapter: *Climate Change, Credit Risk and Financial Stability* by Oguntuase (2020), from the edited volume Banking and Finance, InTech Open Business Report: *Integrating Climate Risk into Credit Risk Modelling* by Reply	SP
10	Incorporating Climate change into a traditional risk management framework	Review of the traditional risk management framework for financial institutions, Role of climate change in skewing the extent of operational risk, How weather-related hazards (intensified by climate change) can result in financial liabilities	Are Bank's Operation Risks Significantly affected by Climate Change? By G. Hopper (2024)	SP

Evaluation Components

✓ Components	✓ Weightage
Quiz	50
End-term	50